Numerical Methods

Practical tasks

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HIT

Initial Value Problem

Given the initial value problem defined by

$$\frac{dy}{dx} = y(1 + x^2), \quad y(0) = 1$$

find the values of y for $x = 0.2,\ 0.6$ and 1.0 using the Euler, the modified Euler and fourth order Runge-Kutta methods. Compare the computed values with the exact values.

Boundary Value Problem

- Solve the boundary value problem defined by

$$y'' - y = 0, \quad y(0) = 0, \quad y(1) = 1,$$

  by finite difference method. Compare the solutions obtained at $y(0.5)$ with the exact value. Take $h = 0.5$ and $h = 0.25$.

- Using Galerkin's method, compute the value of $y(0.5)$ given that

$$y'' + y = x^2, \quad y(0) = 0, \quad y(1) = 0,$$

Eigenvalues

Using The Power Method solve the Eigenvalue problem (find first and, if possible, the second Eigenvalue and Eigenvector)

$$A = \begin{bmatrix} 2 & 11 & 3 & 6 \\ 11 & 5 & 11 & 8 \\ 3 & 12 & 5 & 4 \\ 7 & 5 & 2 & 3 \end{bmatrix}.$$

Using any possible numerical calculation find all Eigenvalues and their Eigenvectors

Heat Equation

Solve the heat equation

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$$

subject to the conditions $u(x,0) = 0 < u(0,t) = 0$ and $u(1,t) = t$. With $h = \frac{1}{4}$ and $l = \frac{1}{16}$, compute the value of $u\left(\frac{1}{2}, \frac{1}{8}\right)$ using Crank-Nicolson formula.

Wave Equation

Solve the equation

$$\frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 u}{\partial x^2}$$

with the conditions $u(0,t) = u(1,t) = 0$, $u(x,0) = \sin \pi x$, and $\frac{\partial u}{\partial t}(x,0) = 0$. Taking $h = 0.25$ and $l = 0.2$, compute $u(0.5, 0.4)$ in two time steps and compare your result with the exact solution given by $u(x,t) = \sin(\pi x) cos(\pi t)$.

Poison Equation

Solve Poisson's equation

$$\nabla^2 u = 8x^2y^2$$

for the square grid shown below $(h = 1)$.